Full Stack Assignment – 45

Assignment ID: FSD-45
Title: Immutable Audit Log Management System
Duration: 3 Hours

1. Objective

Design and develop an **Immutable Audit Log Management System** that records critical system actions performed by users.
The system must ensure audit logs are **append-only**, tamper-proof at the application level, and **persisted in a database**.

2. User Roles

USER

- Register and log in to the system
- Perform allowed actions in the system
- View own activity logs (read-only)

ADMIN

- Register and log in to the system
- View audit logs for all users
- Filter logs based on user, action type, or date range

3. Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use JWT-based authentication
- Protect all backend APIs using JWT
- Enforce role-based access control on APIs and frontend views

Audit Log Generation

- System must automatically generate an audit log entry for:
  - User login
  - User logout
  - Any create or update operation performed by a user
- Audit logs must include:
  - User identifier
  - Action performed
  - Timestamp
- All audit log entries must be stored in the database

Audit Log Access

- USER can view only their own audit logs
- ADMIN can view all audit logs
- Audit logs must be retrievable with basic filtering

4. Business Rule (Critical)

Audit logs must be **immutable**.
Once an audit log entry is created, it must **not be edited or deleted** under any circumstances.

5. Technical Constraints (Mandatory)

- All audit log data must persist in a database
- Hardcoded or in-memory audit storage is not allowed
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6. Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Activity log listing for USER (read-only)
- Global audit log view with filters for ADMIN

UI design is not evaluated; audit integrity and correctness are the priority.

7. Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview

- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Instructions to run the project locally
- Live frontend and backend links

8. Evaluation Note

Evaluation focuses on **immutability enforcement, audit accuracy, role-based visibility, database persistence, and end-to-end execution**.